

SWE 101 – Semester Project Guide Part-2

1. Team Information

- Elisa-02052548
 - Katerina-02052534
 - Isadora-02052528
 - Blerta-02052510
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- Proposed Scrum Role Rotation Plan (who starts as Scrum Master?):
 - Sprint 1: Elisa – Scrum Master
 - Sprint 2: Katerina
 - Sprint 3: Isadora
 - Sprint 4: Blerta

2. Project Title

.SafeNest – Smart Home Security System

3. Problem Statement

Many homeowners lack a reliable and affordable way to monitor their homes in real time. Traditional security systems are often expensive, difficult to access remotely, and do not provide instant alerts. As a result, users may not be aware of security threats until it is too late. This creates a need for a system that provides real-time monitoring, immediate notifications, and remote access to security cameras in a simple and efficient way.

4. Proposed Solution

SafeNest is a smart home security system that allows users to monitor their homes through connected cameras and a mobile application. The system provides live video streaming, motion detection alerts, and secure storage of recordings. Users can access the system remotely and manage multiple cameras from one interface. The system is designed for homeowners and small property owners who need an affordable and easy-to-use security solution.

The system will be developed as a mobile and web-based application using a backend server for processing and data storage.

The system will NOT include advanced AI facial recognition or large-scale enterprise security management features, keeping the scope suitable for a semester project.

5. Stakeholder

Stakeholder	Interest	Risk
Homeowner	Easy monitoring of home	May stop using if UI is complex
Technician	...System maintenance	System downtime issues
Admin	System control	System vulnerabilities
Guest user(optional)	Limited access	Privacy concerns

6. High-Level Requirements

6.1 Functional Requirements (5–8 items)

FR1: The system shall allow users to log in securely

FR2: The system shall allow users to view live camera feeds

FR3: The system shall send motion detection alerts to users

FR4: The system shall allow users to add and remove cameras

FR5: The system shall allow users to view recorded footage

FR6: The system shall support multiple cameras per user

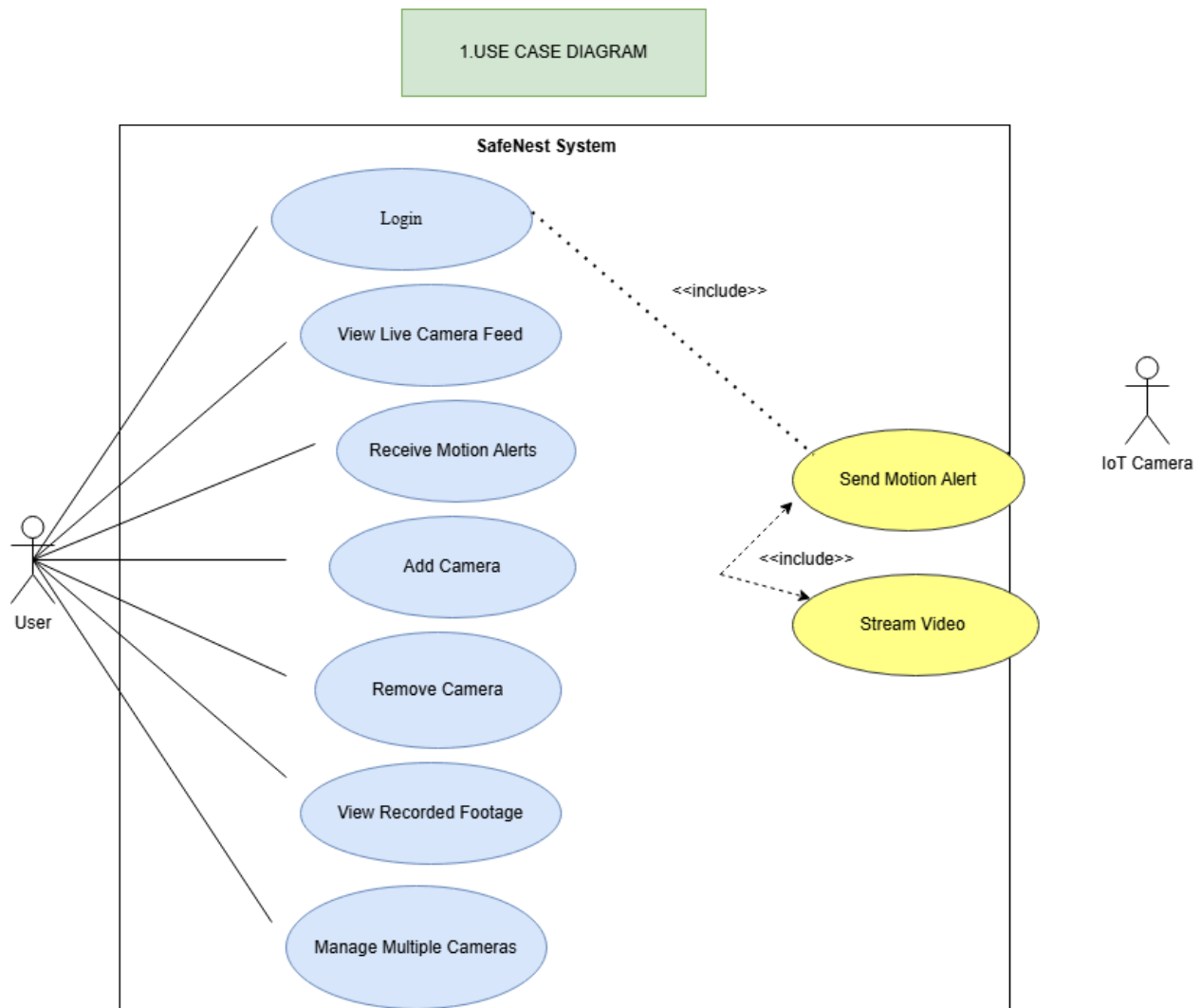
6.2 Non-Functional Requirements (At least 3)

System response time shall be under 2 seconds for normal actions

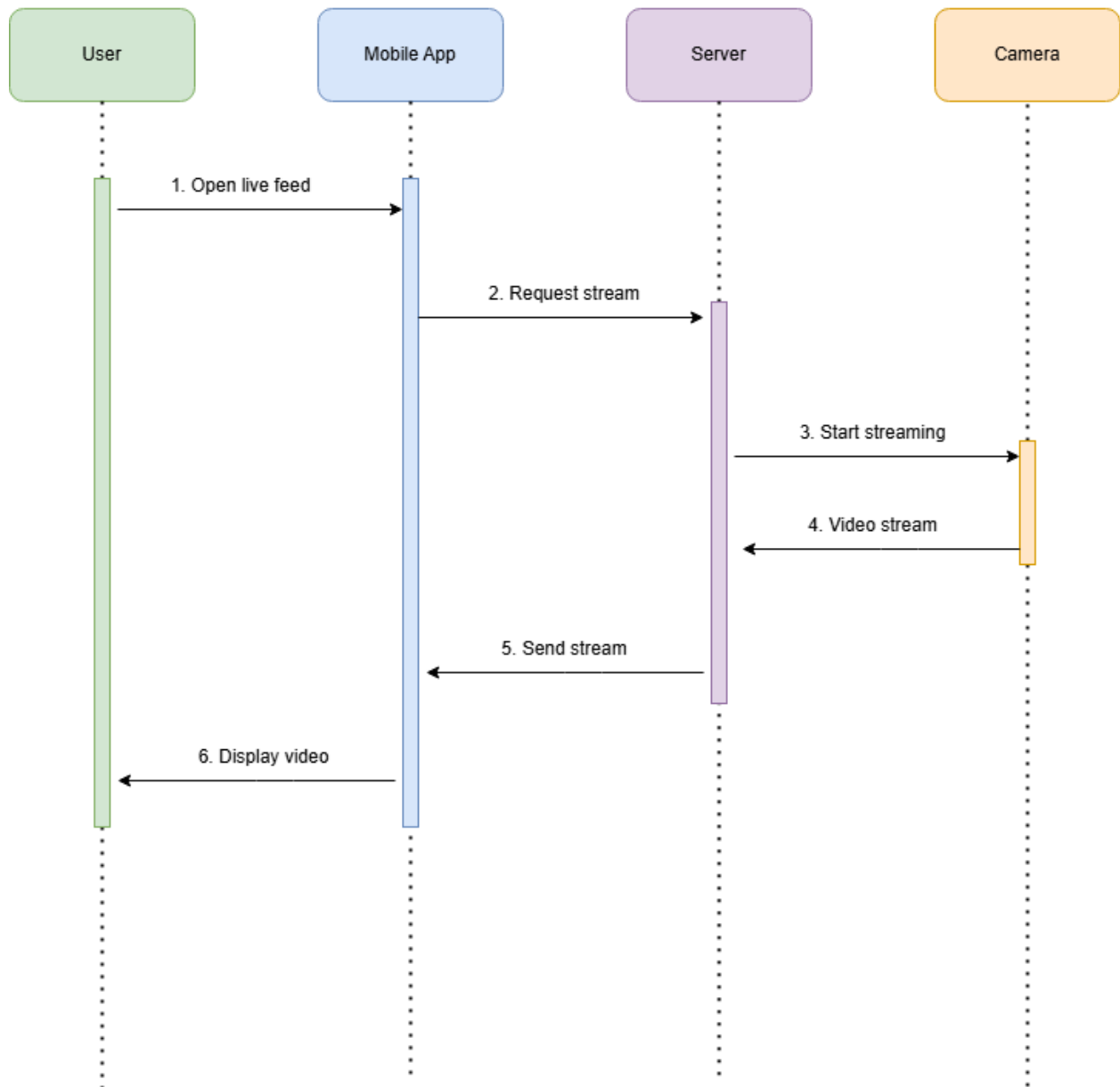
The system shall ensure secure data encryption for user data

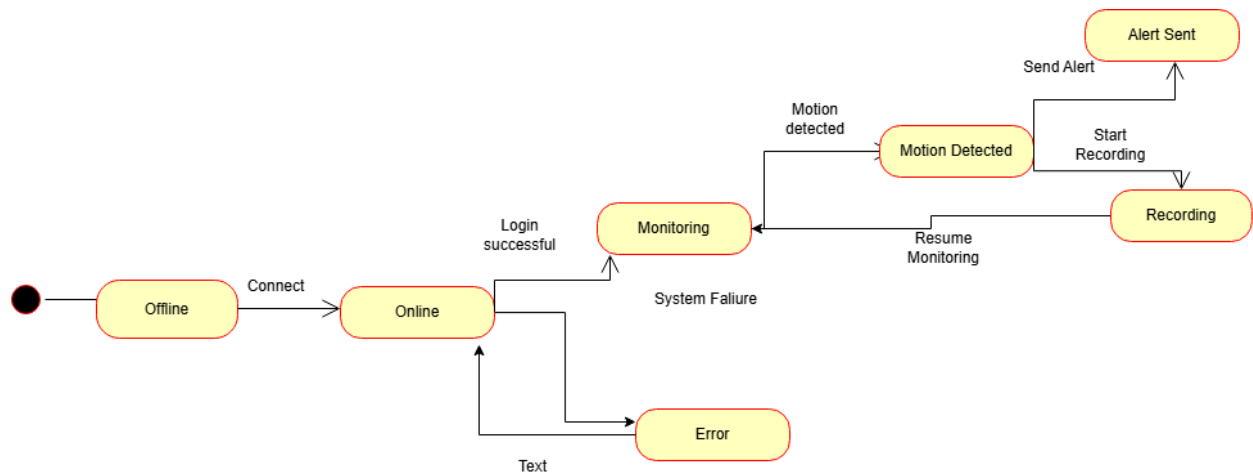
The system shall be available 24/7 with minimal downtime

7. Initial System Model



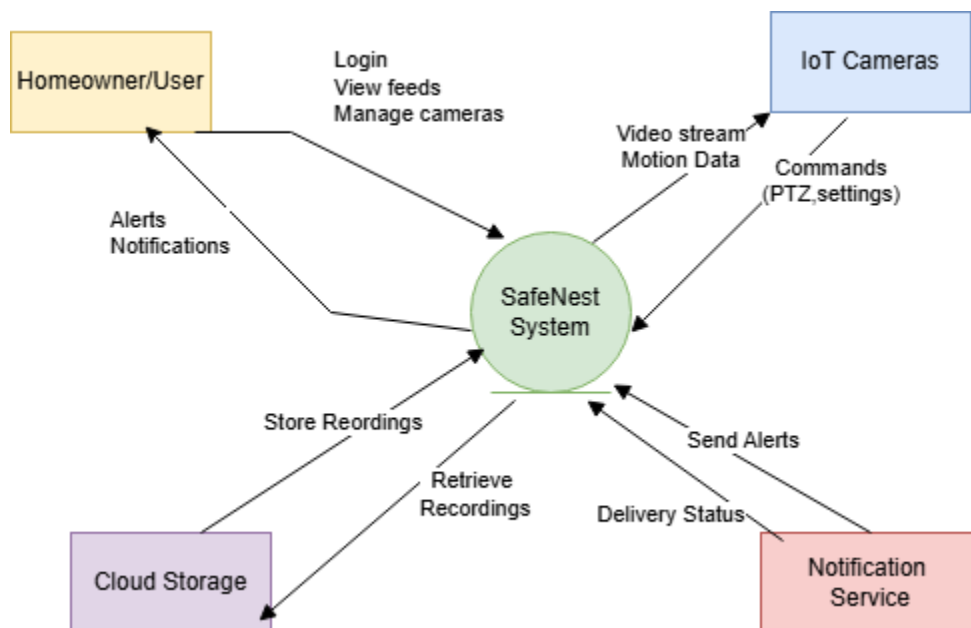
2. SEQUENCE DIAGRAM



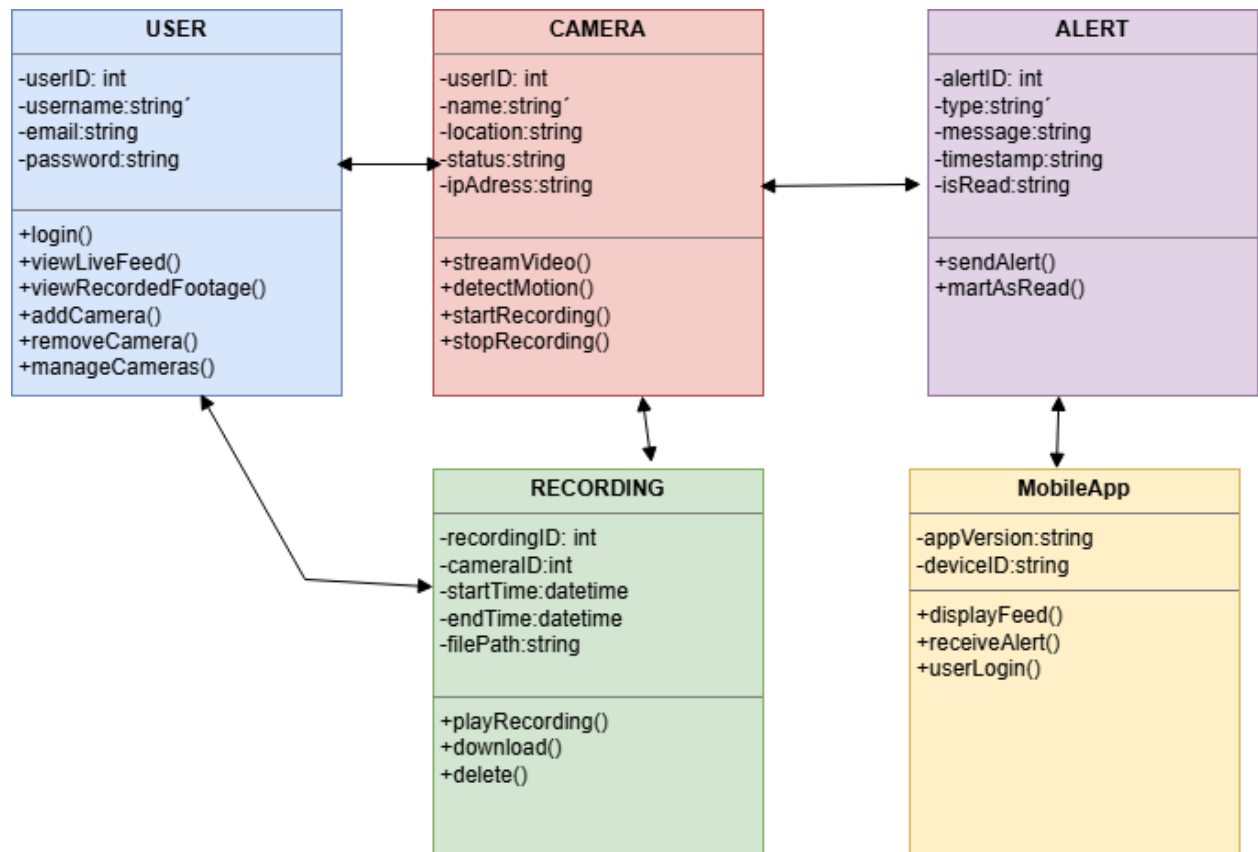


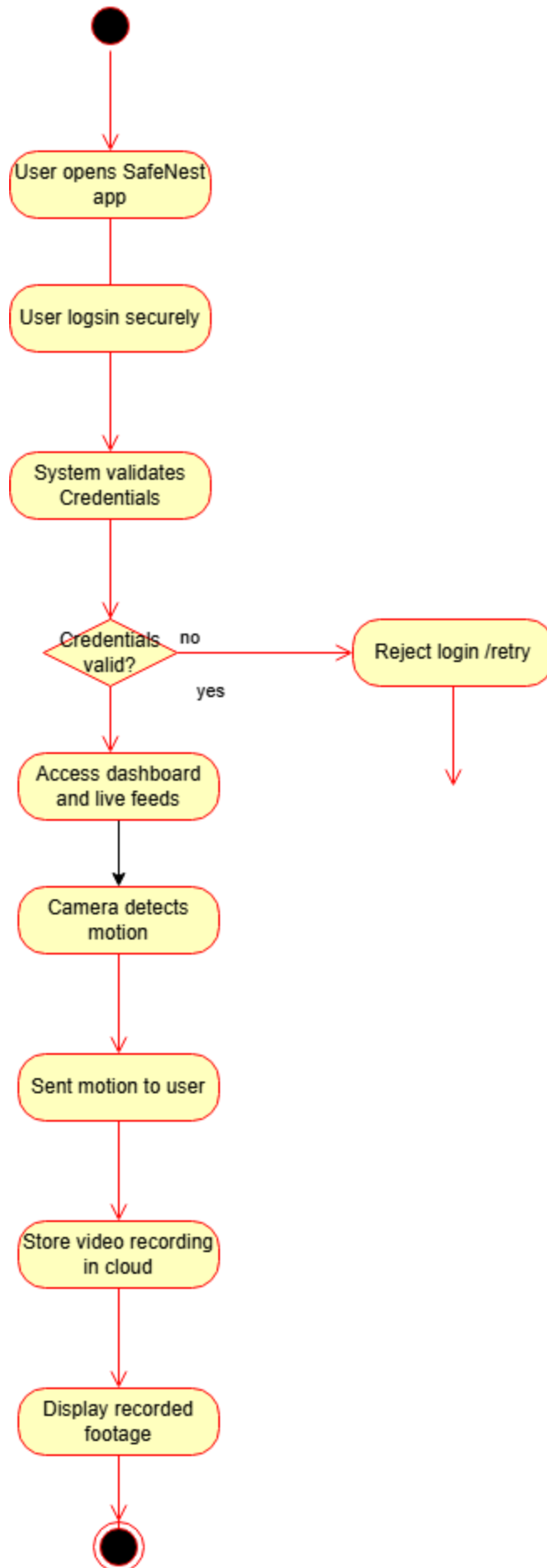
State Machine Diagram

Context Diagram



Class Diagram





Activity Diagram

8. Technology Stack (Tentative)

Programming Language: Java

Framework: Spring Boot

Database: MySQL

Version Control: Git & GitHub

Hosting: Local / Cloud (optional)

Justification:

These technologies are widely used, beginner-friendly, and suitable for building scalable backend systems.

9. Risks (At least 3)

Risk	Likelihood	Mitigation
Motion detection may generate false alerts	Medium	Adjust detection sensitivity and test system regularly
Video storage may become full	High	Implement automatic deletion of old footage
Internet connection failure may interrupt remote access	Medium	Continue local recording during outages
Unauthorized access to video data	Medium	Use secure authentication and data encryption

10. Development Plan (High Level)

Sprint 1: Requirements + system design

Sprint 2: UI mockups + diagrams

Sprint 3: Final report + testing + improvements

11. Ethical & Social Considerations

The system handles personal security data such as camera feeds. Misuse of this data could lead to privacy violations. To prevent this, the system will use secure authentication and encrypted data storage. Access to video feeds will be restricted to authorized users only.